

Tutorial – 03

01. Write an MPI program to find the similarity between two integer value sets with the help of **Dice Coefficient** (**Dice coefficient = $2 * |A \cap B| / (|A| + |B|)$**).

$$\text{Dice Coefficient} = \frac{2 \times |A \cap B|}{|A| + |B|}$$

For example, Consider the 2 sets $A = \{1, 2, 3, 4, 5, 6\}$ and $B = \{1, 3, 5, 7\}$, $A \cap B = \{1, 3, 5\}$ and size of A and B are 6 and 5 respectively. Dice coefficient = $2 * 3 / (6 + 5) = 0.545$. Sets A and B are 54.5% similar.

Initially the process with id 0 has two sequences of integer values:

Sequence 1: 45, 41, 92, 79, 11, 83, 97, 91, 98, 67, 53, 34, 68, 72, 76, 12,
78, 59, 64, 06, 89, 62, 26, 23, 17, 35, 36, 18, 25, 55, 43, 56

Sequence 2: 05, 18, 96, 63, 23, 10, 42, 34, 58, 84, 35, 61, 51, 41, 01, 13,
45, 22, 53, 44, 56, 96, 62, 71, 67, 72, 78, 46, 83, 91, 97, 98

- Master should send the Whole Values of Sequence 1 to all other processes.
- Distribute Sequence 2 among four processes. Each process will receive 8 integer values from sequence 2.
- Each process including process with id 0 should print the received integer values of Sequence 2.
- Each process including process with id 0 should calculate and print $|A \cap B|$.
- Process with id 0 should collect $|A \cap B|$ from all processes and calculate the Dice Coefficient and similarity.

02. Write an MPI program to perform parallel matrix–vector multiplication using only point-to-point communication (**MPI_Send** and **MPI_Recv**).

Initially, the process with **id 0** has the following data **Matrix A** and **Vector V**,

A	=	2	4	6	8	1	3	5	7
		1	3	5	7	2	4	6	8
		9	11	13	15	2	4	6	8
		10	12	14	16	1	3	5	7
		5	7	9	11	3	6	8	2
		6	8	10	12	4	7	9	3
		13	15	17	19	5	8	10	4
		14	16	18	20	6	9	11	5

X	=	1
		2
		3
		4
		5
		6
		7
		8

- Send the whole of vector X to all other processes.
- Distribute the rows of matrix A among eight processes. Each process will receive one row (8 elements) from matrix A.
- Each process including process with id 0 should print the received row of matrix A.
- Each process including process with id 0 should compute and print its partial result that is the dot product of the received row with vector X.

$$Y_i = \sum_{j=0}^7 A[i][j] \times X[j]$$

- Process with id 0 should collect all partial results from other processes and construct the final result vector Y.
- Process with id 0 should print the final result vector Y.